

# Sukhrob Ilyosbekov

Machine Learning Engineer · Computer Vision &amp; LLM Systems

📍 Boston, MA   📞 (857) 867-1942   ✉️ [silyosbekov@gmail.com](mailto:silyosbekov@gmail.com)

 [suxrobgm.net/research](https://suxrobgm.net/research)  [Google Scholar](https://scholar.google.com/)  [linkedin.com/in/suxrobgm](https://linkedin.com/in/suxrobgm)  [github.com/suxrobgm](https://github.com/suxrobgm)

## SUMMARY

Machine-learning engineer with three papers across computer vision, cell biology, and multimodal ML, plus nine years of production software work. Most of it comes down to getting deep-learning and LLM systems out of the notebook and into something people use: imaging models inside hospital workflows, an AI freight dispatcher, multi-model pipelines in a live SaaS product. Applied-math background, and comfortable with the part nobody writes papers about, keeping a model working after it ships.

## TECHNICAL SKILLS

**AI/ML:** PyTorch, TensorFlow, CNNs, Vision Transformers, EfficientNet, YOLO, contrastive learning, GradCAM++ , OpenCV, LLM and agentic systems, RAG, MCP, Claude/GPT/DeepSeek APIs

**Math & Modeling:** Probability and statistics, linear algebra, optimization, regression, time-series forecasting, feature engineering, experiment design and A/B testing, model evaluation and error analysis

**Languages:** Python, C#, C++, TypeScript, Kotlin

**Data & Serving:** NumPy, pandas, scikit-learn, FastAPI, PostgreSQL, MongoDB, Redis, OHIF/DICOM

**Infrastructure:** CUDA, mixed-precision and multi-GPU training, model serving and monitoring, Docker, Kubernetes, GitHub Actions, AWS (S3, Lambda, Cognito), Azure

**Software:** ASP.NET Core, Node.js, Bun, NestJS, React, Next.js, Angular

## EDUCATION

**Northeastern University**, M.S. in Computer Science, GPA 4.0, Boston, MA

*Jan 2024 – May 2026*

**Suffolk University**, B.S. in Computer Science, Cum Laude, Boston, MA

*Sep 2021 – May 2023*

**Honors:** National Physics Olympiad, 1st Place (Uzbekistan) · Associate Member, Sigma Xi · University Coding Competition Winner, TUIT

## PROFESSIONAL EXPERIENCE

Senior AI/ML Engineer | EmTech Care Labs | Portland, ME

*Jan 2025 – Present*

- Built the machine-learning features for a HIPAA-compliant Alzheimer's care platform. The main one is a decline-risk model over daily activity and care-log data for roughly 1,200 residents; it holds 0.81 AUC and flags someone about three weeks before the care team escalates on its own.
- Added an NLP pipeline that reads roughly 500 free-text clinical notes a day and pulls out structured details (symptoms, medications, care events) at about 0.90 F1 on a held-out set. Manual charting time for the care team dropped by roughly a third.
- Stood up the ML infrastructure under HIPAA constraints: de-identified training data, nightly retraining, serving behind the API at sub-200 ms p95, drift monitoring. The platform cleared an outside HIPAA readiness review with no critical findings.

**Machine Learning Engineer** | [AllFactors](#) | Santa Clara, CA

Oct 2021 – Dec 2024

- Built property-valuation and market-forecasting models over roughly 2M listings across 40+ metros. Median absolute valuation error went from about 9% to 5% against the rules-based estimate the product shipped with.
- Designed the real-time feature pipeline (SignalR + SQL Server change tracking) that fed live market data to the models and pushed updated predictions out to hundreds of concurrent users in under a second. The models retrained nightly on the day's transactions.
- Productionized the models into the customer-facing dashboards and added automated tests around the data and prediction flows.

Earlier: Software Engineer at Smart Meal Service, Russia (2020 – 2021), building C#/.NET control software for self-service food kiosks and a robotic cashier. Game Developer at Pentalight Technology, Malaysia (2020 – 2021), multiplayer networking and hand tracking for a VR smart-city simulation in Unity. Freelance Developer (2019 – 2020), Python NLP and .NET applications. Game Developer at EC Dev Team, Uzbekistan (2016 – 2019).

## SELECTED AI/ML PROJECTS

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### MorphoCLIP

[GitHub](#) | [Paper](#)

*PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1*

- Built a text-supervised contrastive model that matches Cell Painting microscopy images of drug- and gene-perturbed cells to descriptions of the treatment in ordinary language. The backbones (DINOv3, BioClinical ModernBERT) stay frozen and only small projection heads train, so it fits on one consumer GPU. Tested on CPJUMP1: 51 plates, 3M+ images. Published on arXiv, joint work with two Northeastern co-authors; I led the design, training, and evaluation.

### Localize, Don't Beautify

<https://arxiv.org/abs/2608.02841>

*Image-editing APIs, inpainting, ArcFace, prompt engineering*

- Ask a commercial image editor to change one feature of a face and it beautifies the whole thing, a problem when the picture is meant to be a surgical preview. Tested how much of that can be fixed client-side, without touching the model: prompt-only steering, masked compositing, and model-based inpainting across six editors. Plain masking localized better. Published on arXiv.

### MelanomaNet

[GitHub](#) | [Paper](#)

*PyTorch, EfficientNet V2, GradCAM++ , OpenCV*

- Trained a deep-learning model to classify skin lesions across nine diagnostic categories on a standard public dermatology dataset of 25K+ images. It reaches 0.86 weighted F1.
- The interesting part is the interpretability. The model shows heatmaps of where it looked, broken down along the same ABCDE criteria (Asymmetry, Border, Color, Diameter, Evolution) dermatologists already work from, so a clinician can check the reasoning. Published on arXiv.

### Med Image Scanner

<https://github.com/suxrobgm/med-image-scanner>

*FastAPI, PyTorch, OpenCV, OHIF Viewer, DICOM, Next.js*

- HIPAA-compliant platform that pulls X-ray, CT, and MRI scans straight from hospital imaging systems and runs PyTorch detection models over them. It flags pneumonia on chest X-rays and intracranial hemorrhage on head CTs.
- Predictions render as overlays inside OHIF, the viewer radiologists already work in. De-identification happens on the fly, with audit logging and role-based access behind it.

### LogisticsX

<https://logisticsx.app> | [GitHub](#)

*Claude API, MCP, .NET 10, EF Core, SignalR, Angular 21, Kotlin KMP, PostgreSQL*

- Built an AI dispatcher that matches freight loads to trucks, checks the driver's federal hours-of-service, and plans the multi-stop route, no human in the loop. Runs on the Claude API through a custom tool-use agent. Assigning a load used to take about fifteen minutes; now it takes seconds.
- Shipped the whole product around it: web portals for dispatchers, customers, and brokers, a cross-platform driver app, and a multi-tenant backend wired into the big freight load boards and tracking providers.

### Bookshelf Scanner

<https://github.com/suxrobgm/bookshelf-scanner>

*YOLO, Moondream2 VLM, FastAPI, Angular*

- Point a camera at a bookshelf and get back a list of what is on it. A YOLO segmentation model cuts out each spine, then a vision-language model reads the title and author off it, with a web app for fixing misreads and exporting the library.

### OGStack

<https://ogstack.dev> | [GitHub](#)

*Next.js 16, ElysiaJS, Bun, Satori, Cloudflare R2, Stripe*

- Built the multi-model layer that routes work across Claude, GPT, and DeepSeek to read a web page and pick up its branding, then generates the social-share image, backgrounds included. Runs as a SaaS serving cached images off a global CDN; 10K+ images during the beta.